

SECOND TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT) — THIRD SEMESTER, NOVEMBER 2024

Paper Code: ES-201

Subject: Computational Methods

Time: 1½ Hrs.

Max. Marks: 30

Note: Attempt Q. No. 1, which is compulsory, and any two more questions from remaining.

Q. No.	Question	Marks	CO
1(a)	Investigate the value of λ so that the system of equations: $3x - y + 4z = 3$, $x + 2y - 3z = -2$, $6x + 5y + \lambda z = -3$ have (i) unique solution, (ii) an infinite number of solutions.	[2.5]	CO3
1(b)	Define a first degree spline function and check whether $f(x)$ is a first degree spline function or not: $f(x) = \{ x, -1 \leq x < 0; 2 - x, 0 \leq x < 1; 2x - 1, 1 \leq x \leq 2 \}$	[2.5]	CO3
1(c)	Using Picard's method, find the first approximation for the solution of differential equation $dy/dx = 1 + xy$, $y(0) = 1$ for any x .	[2.5]	CO4
1(d)	Using Euler's method, compute $y(0.1)$ from $dy/dx = x + y + xy$ with the initial condition $y(0) = 1$ in two steps.	[2.5]	CO4
2(a)	Solve the system of equations using Gauss Jordan method: $2x - 6y + 8z = 24$, $5x + 4y - 3z = 2$, $3x + y + 2z = 16$	[5]	CO3
2(b)	Find by Taylor's series method, the approximate value of y at $x = 0.1$ correct to five places of decimals from the differential equation $dy/dx = x^2y - 1$, $y(0) = 1$.	[5]	CO4
3(a)	Determine the Cholesky factorization of the matrix: $A = \begin{bmatrix} 4 & 2 & 14 \\ 2 & 17 & -5 \\ 14 & -5 & 83 \end{bmatrix}$	[5]	CO3
3(b)	Using Runge-Kutta method of fourth order, solve $dy/dx = x^2 + y^2$, $y(0) = 1$. Take step size $h = 0.1$ and determine approximations to $y(0.1)$ and $y(0.2)$ correct to two decimal places.	[5]	CO4
4(a)	Determine the numerically dominant eigenvalue and eigenvector of the matrix: $A = \begin{bmatrix} 3 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 3 \end{bmatrix}$ using Power method with initial vector $x^{(0)} = [1, 0, 1]^T$.	[5]	CO3
4(b)	Solve the partial differential equation $\partial u / \partial t = \partial^2 u / \partial x^2$ subject to the conditions $u(0, t) = u(1, t) = 0$ and $u(x, 0) = \sin(\pi x)$, $0 \leq x \leq 1$ using explicit method for $h = 1/3$ and $k = 1/36$ up to two levels only.	[5]	CO4

**** End of Question Paper ****